SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE
Constituent of Symbiosis International (Deemed University), Pune

Course	Digital Signal Processing Lab	**Experiment No.**	11
Name	Dewashish Lambore	**PRN**	_____
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FIR Filter Design by Windowing

Heart rate and breathing rate from one fingertip pulse sensor

Abstract: Linear-phase FIR filters were designed by the window method, from the ideal sinc impulse response and the rectangular, Hann, Hamming, Blackman and Kaiser windows, and implemented as non-recursive difference equations. For the same specification the four classical windows gave stopband attenuations of -21, -44, -52, -75 dB, matching the textbook values, at the cost of progressively wider transition bands; Kaiser's formulas sized a 60 dB design at N = 1511 taps (59.4 dB achieved). The filters were applied to an 8-minute photoplethysmogram (PPG) from an intensive-care patient (PhysioNet BIDMC): a 251-tap band-pass filter (0.5 to 4 Hz) recovered the heart rate with a mean absolute error of 1.49 bpm against the bedside monitor, and a 501-tap low-pass filter (0.5 Hz) recovered the breathing rate within 1.33 breaths per minute of clinicians' manual annotations. The filters' constant group delay (1 s and 2 s) was removed exactly, and longer filters improved rejection but could not remove in-band motion artefacts.

Keywords: *FIR filter, window method, Hamming, Blackman, Kaiser window, linear phase, group delay, difference equation, photoplethysmography, heart rate, respiration rate, Python*

1. Aim and Objectives

Aim: to design FIR filters by the window method, implement them by their difference equation, plot their frequency response, and use them to extract physiological information from a real signal, in Python.

- Derive windowed-sinc low-pass and band-pass filters and compare five windows by ripple, attenuation and transition width; size a filter with Kaiser's formulas.
- Implement the filters as $y[n] = \Sigma\, b_k\, x[n - k]$, verify linear phase and constant group delay.
- Extract heart rate and breathing rate from a PPG and validate them against hospital reference data.

2. Real-World Application

A **photoplethysmogram** (PPG) is measured by shining light into the skin and recording how much is absorbed as blood volume changes with each heartbeat. The same sensor is in hospital pulse oximeters, smartwatches and fitness bands. Its raw signal mixes three things at different frequencies: the cardiac pulse (about 1 to 3 Hz), a slow modulation by breathing (about 0.1 to 0.5 Hz), and noise from motion and electronics at higher frequencies. Separating them is a filtering problem, and FIR filters are preferred because their **linear phase** does not distort the pulse shape or shift beats relative to each other, which matters for heart-rate variability and clinical waveform analysis. Table 1 lists other uses of window-designed FIR filters.

Table 1. Where window-designed FIR filters are used

Field	Filter	Why FIR / linear phase
Wearables, pulse oximeters (this experiment)	band-pass and low-pass on PPG	keeps pulse shape and beat timing
ECG and EEG instruments	band-pass 0.5 to 40 Hz, notch	no phase distortion of clinical waveforms
Audio crossovers, equalisers	low-pass / high-pass pairs	phase-coherent summation of loudspeakers
Digital communications	pulse shaping, matched filters	symmetric taps avoid inter-symbol distortion
Seismology	band-pass of seismograms	arrival times of waves are preserved

3. Theory

An ideal low-pass filter with cut-off f_c has an infinitely long sinc impulse response. The **window method** delays it by $M/2$ (with $M = N - 1$) to make it causal and multiplies it by a finite window $w[n]$ of N samples:

$$h[n] = h_d[n]\, w[n], \qquad h_d[n] = 2\frac{f_c}{f_s}\,\mathrm{sinc}(2\frac{f_c}{f_s}(n - \frac{M}{2})), \quad 0 \le n \le M \tag{1}$$

A band-pass is the difference of two ideal low-passes, $h_d = h_{d,f_2} - h_{d,f_1}$. The four classical windows are raised cosines; Kaiser's window uses the modified Bessel function I_0 and a shape parameter β:

$$w_{\text{Hamming}}[n] = 0.54 - 0.46\cos\frac{2\pi n}{M}, \qquad w_{\text{Kaiser}}[n] = \frac{I_0(\beta\sqrt{1-(2n/M-1)^2})}{I_0(\beta)} \tag{2}$$

Multiplying by the window convolves the ideal response with the window's spectrum: the **main lobe** sets the transition width and the **sidelobes** set the stopband attenuation (Table 2). For a required attenuation A dB and transition width Δω (rad/sample), Kaiser's formulas give the design directly:

$$\beta = 0.1102\,(A - 8.7)\ \ (A > 50), \qquad N \approx \frac{A-8}{2.285\,\Delta\omega} + 1 \tag{3}$$

Table 2. Window properties (Oppenheim and Schafer, Table 7.2) and measured attenuation

Window	Peak sidelobe	Filter stopband	Transition width Δf	Measured here (N = 251)
Rectangular	-13 dB	-21 dB	≈ 2·fs/M	-21.3 dB
Hann	-31 dB	-44 dB	≈ 4·fs/M	-44.0 dB
Hamming	-41 dB	-53 dB	≈ 4·fs/M	-52.4 dB
Blackman	-57 dB	-74 dB	≈ 6·fs/M	-75.4 dB

The filter is realised by the **non-recursive difference equation**; because the coefficients are symmetric, $h[n] = h[M - n]$, the phase is linear and every frequency is delayed by the same group delay:

$$y[n] = \sum_{k=0}^{M} h[k]\, x[n-k], \qquad H(e^{j\omega}) = A(\omega)\, e^{-j\omega M/2}, \qquad \tau = \frac{M}{2}\ \text{samples} \tag{4}$$

> **Note.** A constant delay is harmless: it can be removed exactly by shifting the output M/2 samples earlier, which is done in this experiment. An IIR filter of similar selectivity (Experiment 12) needs far fewer coefficients but has a frequency-dependent delay that distorts waveform shape.

4. Dataset Description

The **BIDMC PPG and Respiration Dataset** (Pimentel et al., PhysioNet, ODC Attribution licence) contains 53 eight-minute recordings from adult patients at the Beth Israel Deaconess Medical Center, extracted from the MIMIC-II waveform database. Each record has the PPG, impedance respiration and ECG at 125 Hz, the bedside monitor's numeric values once per second, and every breath annotated manually by two trained annotators. Record **bidmc01** (88-year-old male, medical ICU) was used: the binary WFDB file is read directly with its header gains, about 596 kB in total.

Table 3. Dataset attributes

Property	Value	Property	Value
Source	PhysioNet BIDMC 1.0.0	Record	bidmc01, 5 signals
Sampling rate	125 Hz	Length	60,001 samples (8.0 min)
Signal used	PLETH (PPG), unit NU	Gain / baseline	138229 / -63710
Reference heart rate	monitor PULSE, mean 91.0 bpm	Reference breathing	170 annotated breaths
Monitor RESP	mean 21.4 per min	Missing PULSE values	13 of 481 s

Table 4. Dataset preview with the unit conversion

n	Time (s)	Raw int16	PPG = (raw − baseline)/gain	Monitor PULSE at that second
12,500	100.000	7769	0.51710	91
12,501	100.008	7769	0.51710	91
12,502	100.016	7769	0.51710	91
12,503	100.024	7769	0.51710	91
12,504	100.032	7769	0.51710	91
12,505	100.040	7904	0.51808	91

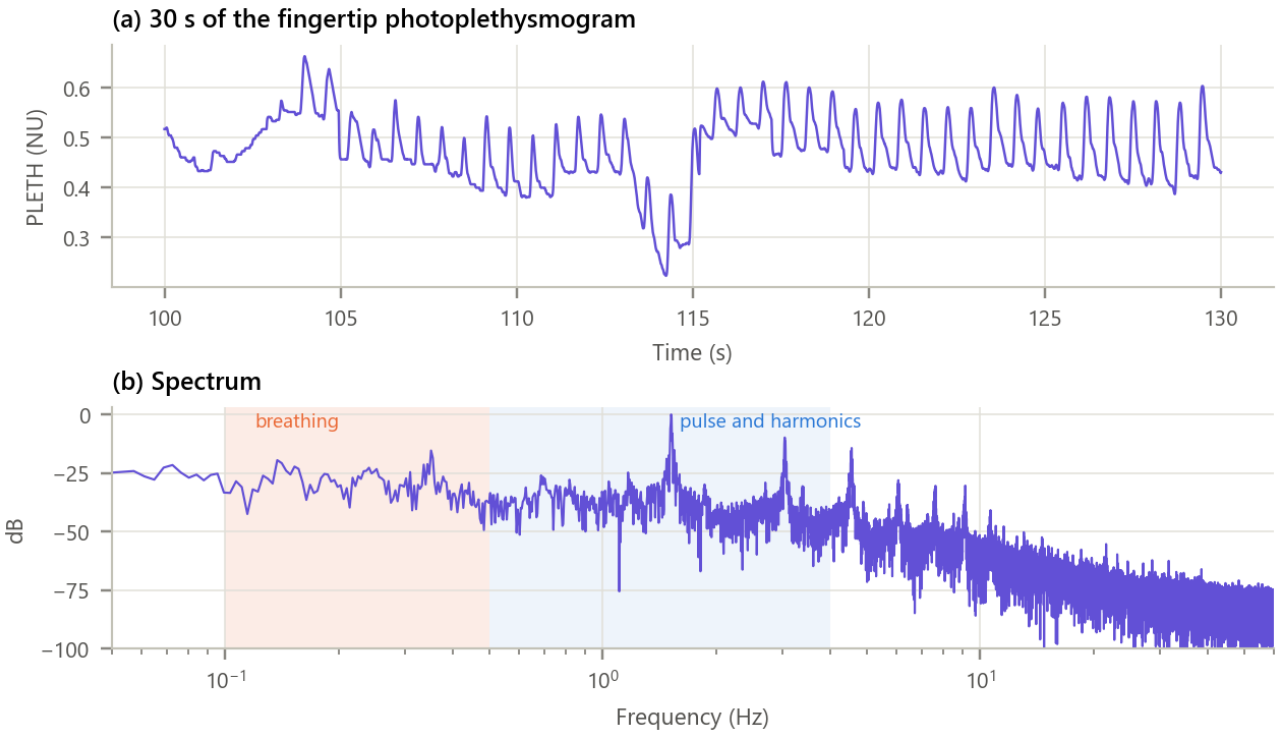


Figure 1. The input photoplethysmogram. (a) 30 s of the signal; (b) its spectrum with the breathing and pulse bands shaded.

Inference. Individual pulses are clearly visible, riding on a slow wave that follows breathing, and a large dip at about 113 s is a motion artefact. The spectrum shows the cardiac fundamental at 1.52 Hz (91 bpm) with harmonics at 3.0 and 4.5 Hz, broad energy below 0.5 Hz from breathing, and a noise floor above 10 Hz. The pulse and breathing bands do not overlap, so linear filters can separate them, provided their transition bands are narrow enough.

5. Methodology and Implementation

The band-pass filter for the pulse uses 251 taps (Hamming), the low-pass for breathing 501 taps (Hamming). The output is advanced by M/2 to remove the delay; beats and breaths are found as peaks at least 0.33 s and 1.5 s apart; rates are averaged in 16 s and 60 s windows and compared with the references.

Table 5. Software modules

Module	Responsibility
`config.py`	Record, file URLs, band edges, tap counts, windows, rate-estimation settings
`data_loader.py`	Download once, parse the WFDB header, read format-16 samples, load references
`fir.py`	Windows, ideal responses, windowed designs, Kaiser formulas, difference equation, H(f), delay
`analysis.py`	Window study, filters, delay compensation, beat and breath detection, validation, tap study
`plots.py` / `gui.py`	Report figures / interactive FIR filter designer
`test_fir.py`	9 unit tests (match with SciPy firwin, symmetry, constant delay, Kaiser spec): all pass

Core of fir.py

```python
def ideal_lowpass(fc, fs, N):                # h_d[n] = 2fc/fs sinc(2fc/fs (n - M/2))
    n = np.arange(N) - (N - 1) / 2
    return 2 * fc / fs * np.sinc(2 * fc / fs * n)

def design_bandpass(f1, f2, fs, N, win="hamming"):
    h = (ideal_lowpass(f2, fs, N) - ideal_lowpass(f1, fs, N)) * window(win, N)
    return h / abs(np.sum(h * np.exp(-2j * np.pi * np.sqrt(f1 * f2) / fs * np.arange(N))))

def fir_filter(b, x):                         # y[n] = sum_k b_k x[n - k]
    y = np.zeros(len(x))
    for k, bk in enumerate(b):
        y[k:] += bk * x[:len(x) - k]
    return y

```

GUI. `gui.py` (Figure 2) designs a low-pass or band-pass filter with any cut-offs, any odd number of taps and any window (including Kaiser with adjustable β). It shows the impulse response with its delay, the magnitude response, the filtered PPG, and the heart or breathing rate with its error against the hospital reference; the coefficients can be saved. Run `python run_experiment.py` and `python gui.py`.

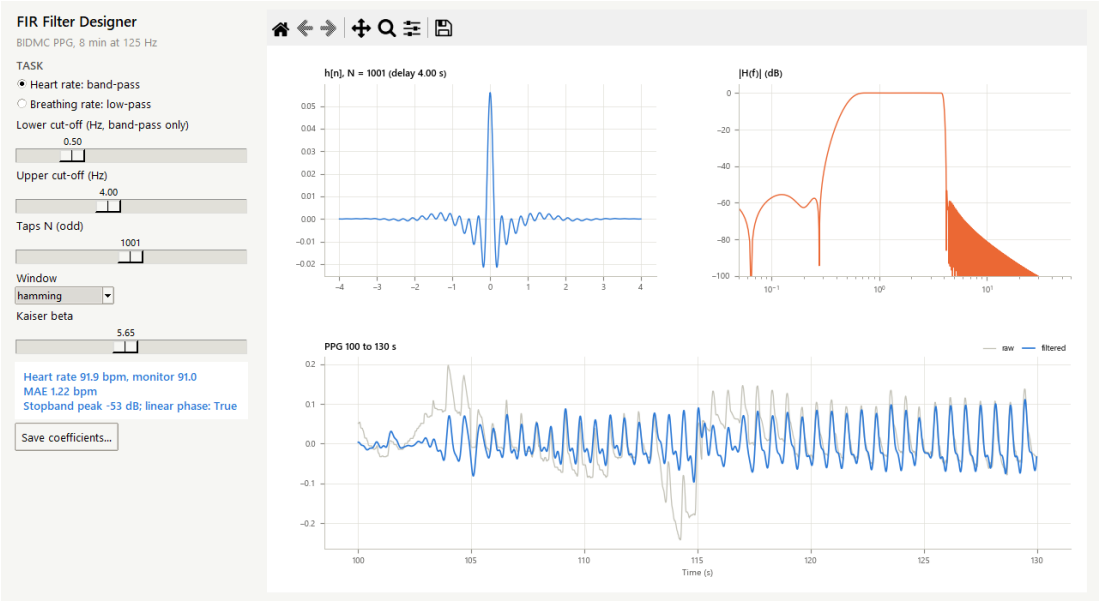


Figure 2. The FIR filter designer with a 1,001-tap Hamming band-pass filter. Stopband better than −50 dB and no baseline wander in the filtered pulse.

6. Calculations

5-tap low-pass by hand. fc = fs/4 (31.25 Hz), N = 5, M = 4, so 2fc/fs = ½ and $h_d[n] = \frac{1}{2}\text{sinc}(\frac{n-2}{2})$:

Table 6. Window-method design of a 5-tap filter

n	n − M/2	h_d[n] = ½ sinc((n − 2)/2)	Hamming w[n]	h[n] = h_d w	normalised h[n]
0	-2	0.0000	0.08	0.0000	0.0000
1	-1	0.3183	0.54	0.1719	0.2037
2	+0	0.5000	1.00	0.5000	0.5926
3	+1	0.3183	0.54	0.1719	0.2037
4	+2	0.0000	0.08	0.0000	0.0000

Sum of h[n] = 0.8438, so dividing by it gives unit DC gain; the taps are symmetric, hence linear phase with a delay of M/2 = 2 samples (16 ms). At fc the normalised filter gives |H| = 0.593 (-4.5 dB), and at fs/2 it gives 0.185. **Kaiser sizing:** for A = 60 dB and a transition of 0.3 Hz, $\Delta\omega = 2\pi \cdot 0.3/125 = 0.01508$ rad, $\beta = 0.1102 \cdot (60 - 8.7) = 5.653$ and $N = (60 - 8)/(2.285 \cdot \Delta\omega) + 1 \rightarrow$ **1511 taps**. **Delays used:** (251 − 1)/2 = 125 samples = 1 s and (501 − 1)/2 = 250 samples = 2 s. The program reproduces these values (checked by `test_fir.py`).

7. Results and Discussion

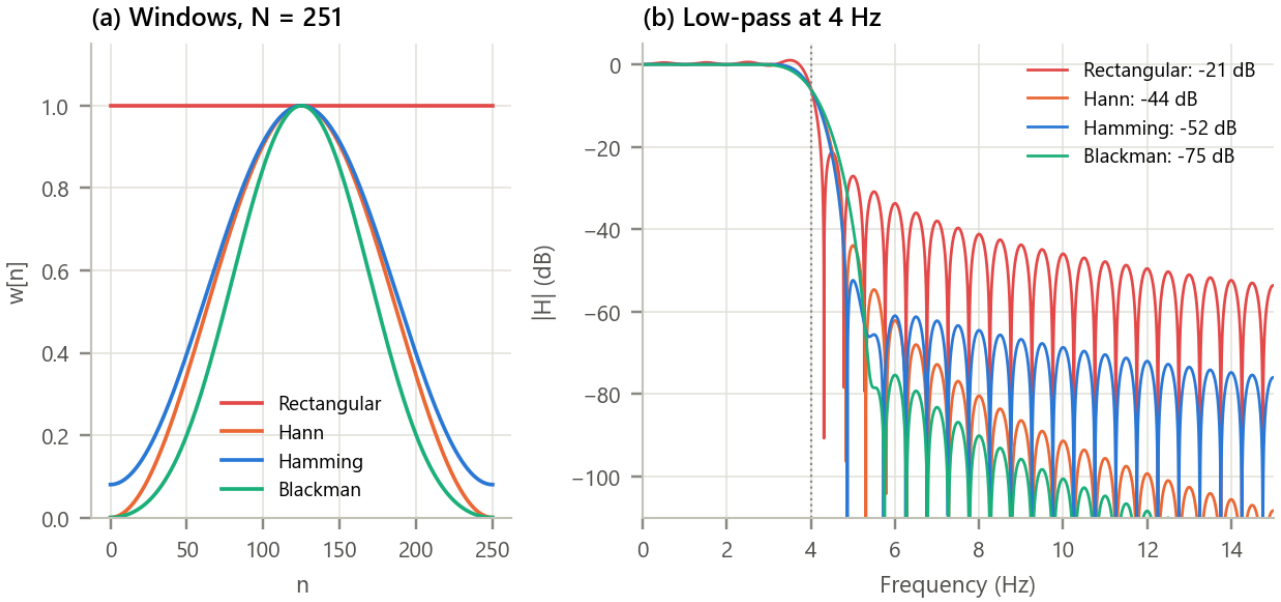


Figure 3. Effect of the window on a low-pass filter. (a) The four windows for N = 251; (b) the resulting low-pass responses at 4 Hz with their measured stopband attenuation.

Inference. Each window trades transition width for attenuation exactly as theory predicts: the rectangular window gives the sharpest edge (−3 to −40 dB in 0.40 Hz) but only -21 dB of attenuation with ripples that decay slowly; Hann reaches -44 dB, Hamming -52 dB and Blackman -75 dB, while the transition widens to 1.27 Hz. Passband ripple falls from 1.30 dB to 0.003 dB. Hamming is a good default: about 53 dB with a moderate transition.

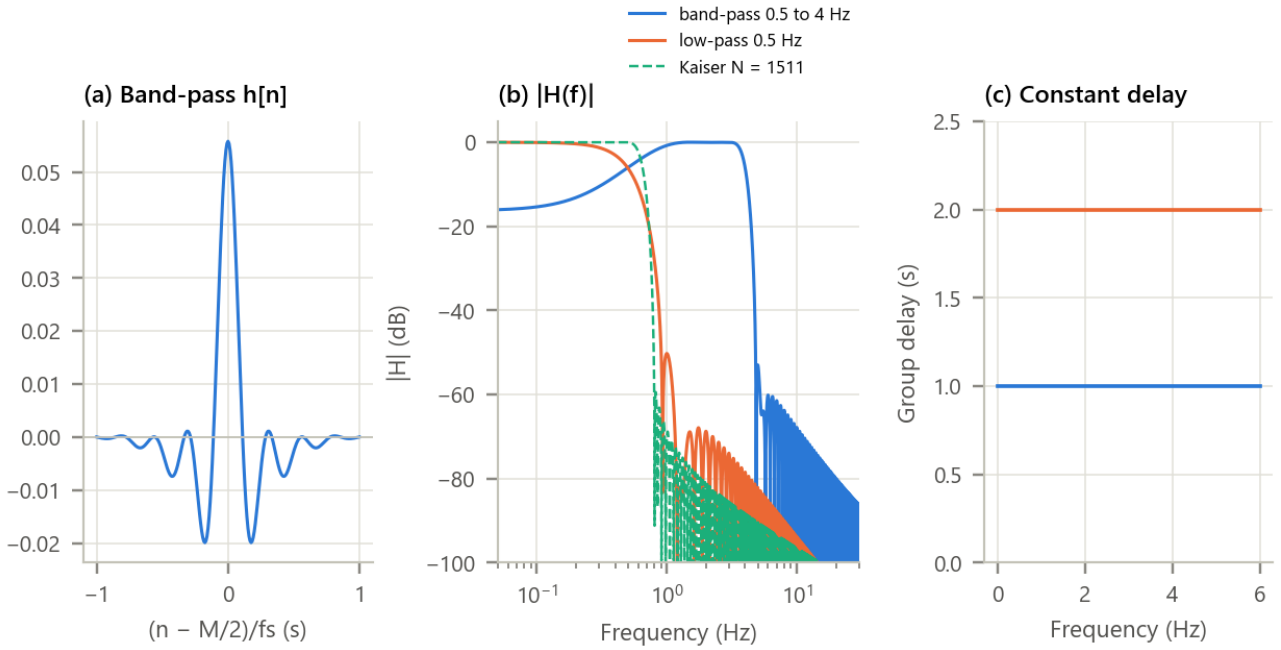


Figure 4. The designed filters. (a) Band-pass impulse response; (b) magnitude responses, including the Kaiser design; (c) group delay.

Inference. The band-pass passes the pulse with 0.01 dB at 1.5 Hz and attenuates 10 Hz noise by 68 dB, but with 251 taps its lower edge is soft: breathing at 0.25 Hz is cut by only 12 dB. The low-pass keeps 0.33 Hz breathing (-1.7 dB) and removes the 1.5 Hz pulse by 69 dB. The Kaiser design reaches 59.4 dB against its 60 dB target with N = 1511 (the formulas are approximations, so a few more taps would close the gap), and shows that a 0.3 Hz transition costs many taps at 125 Hz. Both filters are exactly symmetric (asymmetry 5e-18) and their group delay is flat at 1 s and 2 s (spread 2e-14 samples).

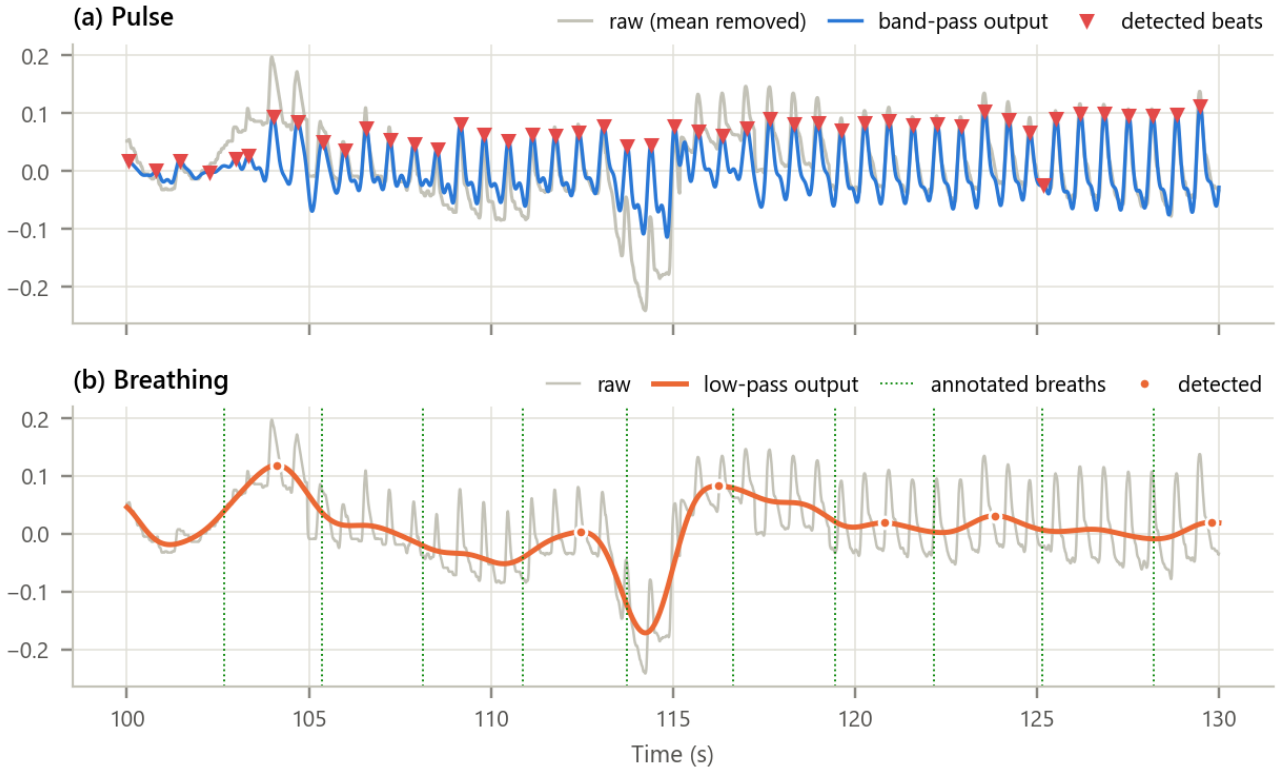


Figure 5. The filters applied to the PPG (100 to 130 s). (a) Raw signal, band-pass output and detected beats; (b) low-pass output, detected breaths and the manual breath annotations.

Inference. After delay compensation the band-pass output lines up beat for beat with the raw pulses: linear phase shifted nothing. Each pulse gives one clean peak except during the motion artefact around 113 s, where the residual slow wander causes a few missed or extra beats. The low-pass output is a smooth breathing wave whose peaks fall close to the annotators' marks, again with trouble around the artefact.

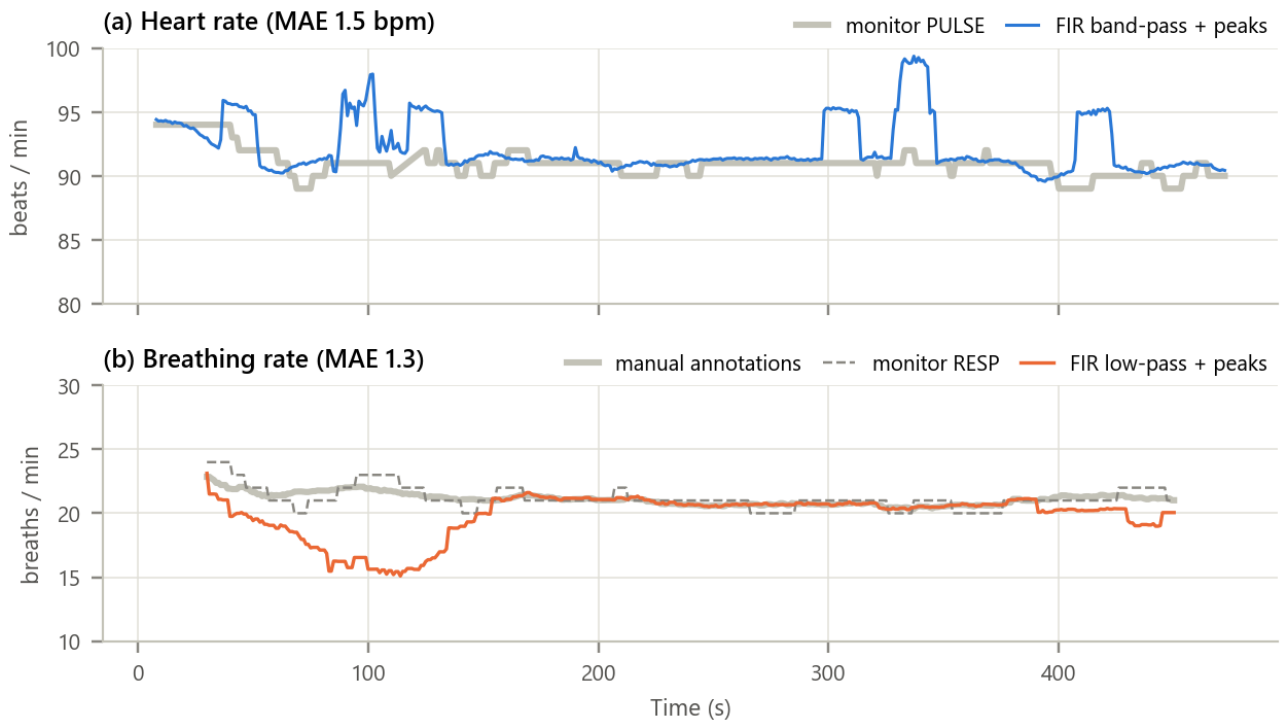


Figure 6. Rates estimated from the filtered PPG against the hospital references. (a) Heart rate (16 s windows) against the monitor; (b) breathing rate (60 s windows) against the manual annotations and the monitor's impedance-based value.

Inference.

- Heart rate: 92.2 bpm against 91.0 bpm from the monitor, mean absolute error 1.49 bpm from 725 detected beats. Short excursions to about 99 bpm coincide with motion artefacts.
- Breathing rate: 19.8 against 21.1 breaths per minute from the annotations (MAE 1.33); 87 % of annotated breaths were found within 1.5 s. The underestimate between 60 and 150 s is the artefact region, where breaths are masked.
- Lengthening the band-pass from 251 to 1001 taps improved the 0.25 Hz rejection from 12 to 58 dB and lowered the heart-rate error from 1.49 to 1.22 bpm, but the artefact spikes (99 bpm) remained: noise that lies inside the passband cannot be removed by any linear filter.

Table 7. Effect of filter length on heart-rate accuracy

Band-pass taps N	Delay (s)	Gain at 0.25 Hz (dB)	HR MAE (bpm)	Windows within ±2 bpm
251	1	-11.9	1.49	79 %
501	2	-22.2	1.22	85 %
751	3	-40.7	1.22	85 %
1001	4	-57.5	1.22	85 %

8. Conclusion

Window-designed FIR filters were built from first principles, matched SciPy's implementation exactly, and showed the predicted trade-off between transition width and stopband attenuation for each window, while Kaiser's formulas sized a filter close to a stated specification. Implemented as a symmetric difference equation, every filter had exactly linear phase and a constant delay that was removed by a simple shift, so the

pulse shape and beat timing were preserved. On a real ICU recording the band-pass filter recovered the heart rate to 1.5 bpm and the low-pass filter the breathing rate to 1.3 breaths per minute of the hospital references, which is the principle behind smartwatch heart and breathing measurement. The price of FIR filters is length: sharp edges at low frequencies need hundreds or thousands of taps and seconds of delay, and no filter, however long, removes artefacts that share the signal's band.

9. Learning Outcomes

- Design low-pass and band-pass FIR filters by the window method and size them with Kaiser's formulas.
- Relate a window's main lobe and sidelobes to the filter's transition width and stopband attenuation.
- Implement an FIR filter as a difference equation, verify linear phase and compensate its group delay.
- Apply FIR filters to a physiological signal and validate the extracted rates against reference data.

10. References

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